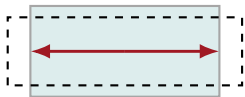


A quartz blank can vibrate in any of these. The cut of the blank and the circuit around it pick one, and that choice, not the quartz, is what sets the frequency



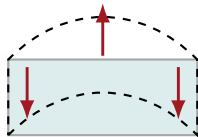
Longitudinal

the bar stretches  
and shortens  
tens of kHz



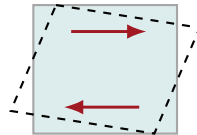
Thickness shear

faces slide past  
each other  
1 to 200 MHz



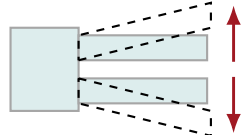
Flexural

the bar bends  
like a diving board  
tens of kHz



Face shear

the square racks  
over into a rhombus  
hundreds of kHz



Tuning fork

two tines swing  
apart and together  
32.768 kHz

The two in the green boxes are the ones you will meet on a circuit board: a thickness shear blank for the megahertz reference, and a tuning fork for the 32.768 kHz real-time clock